CENG 201 – Task 4 Report

Topic: Integrated Hospital System Implementation

Time Complexity Analysis:

In this task, an integrated hospital management system was implemented that combines all the data structures from previous tasks: PatientList (linked list), TreatmentQueue (queue), and DischargeStack (stack). Additionally, a HashMap is used for efficient patient lookup.

The addPatient method adds a new patient to both the PatientList and the patientMap (HashMap). Adding to the PatientList has O(1) complexity as it adds to the head, and adding to the HashMap also has O(1) average case complexity. Therefore, the overall time complexity is O(1).

The addTreatmentRequest method adds a treatment request to either the priority queue or normal queue based on the request's priority flag. Since enqueue operations are O(1), this method also has O(1) time complexity.

The processTreatment method processes the next treatment request by first checking the priority queue, then the normal queue if the priority queue is empty. Both dequeue operations are O(1). It then looks up the patient in the HashMap (O(1) average case), pushes a discharge record to the stack (O(1)), and removes the patient from the list (O(n) in worst case). The overall complexity is dominated by the removePatient operation, resulting in O(n) time complexity.

The sortPatientsBySeverity method uses Bubble Sort algorithm to sort patients by their severity level in descending order. Bubble Sort has a time complexity of $O(n^2)$ in the worst and average cases, where n is the number of patients. The algorithm compares adjacent elements and swaps them if they are in the wrong order, repeating this process until the array is sorted.

System Integration:

The system integrates multiple data structures to efficiently manage different aspects of hospital operations:

- PatientList: Manages admitted patients using a linked list

- TreatmentQueue (Priority and Normal): Manages treatment requests using queues

- DischargeStack: Manages discharge records using a stack

- HashMap: Provides O(1) average case lookup for patients by ID

This integrated approach allows for efficient management of the hospital system, with each data structure optimized for its specific use case. The combination of these structures provides a comprehensive solution for patient admission, treatment processing, and discharge management.